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RESEARCH ARTICLE

Mathematical Model and Algorithm for Accurate Main Content Extraction From News Websites

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ABSTRACT Irrelevant elements like ads, menus, and footers in web pages hinder data extraction and reduce the performance of Retrieval-Augmented Generation (RAG) systems in Large Language Models (LLMs). This paper tackles the challenge of accurately identifying and extracting the main content from web pages to enhance the efficiency of these systems. We present a novel mathematical model and algorithm that leverages the Document Object Model (DOM) structure, effectively isolating relevant content with high accuracy. Our approach is language-neutral and performs well across diverse languages, including those with complex tokenization, such as Arabic. To validate the model, we created a dataset from 500 websites, allowing for comprehensive evaluation and benchmarking. The algorithm's practical application demonstrates a reduction in token usage for LLM tasks, contributing to cost-effectiveness. This work introduces a robust, open-source tool for the academic and commercial communities, fostering further innovation in web content extraction and information retrieval.

INDEX TERMS Information extraction, document object model (DOM), retrieval-augmented generation (RAG), large language models (LLM), main content detection.

I. INTRODUCTION

Web pages often present relevant and irrelevant information, depending on the user's intent. Accurately extracting meaningful content from web pages is crucial for human users and automated systems. However, primary content is frequently embedded among noisy elements such as advertisements, menus, headers, footers, and banners. Removing these noisy elements is essential for effective data extraction. The process of identifying the main content of a web page involves isolating the core body of information by filtering out elements that do not contain valuable information. For example, Figure 1 illustrates the main content of a CNN news article, marked by a red rectangle. The tremendous diversity of web pages, even those created using similar content management systems (CMS), complicates the task of

identifying content sections such as menus, main content, and templates [1], [2], [3].

Website templates are crucial in web development because they provide a reusable framework for crafting new web pages with consistent design. These reusable components enable the automation of web page creation, resulting in uniform and intuitive designs for users. Most websites today are managed through automated systems that generate pages from templates, ensuring a cohesive visual and functional experience for users.

Identifying the main content has become increasingly important as modern Retrieval-Augmented Generation (RAG) systems, such as Langchain, rely on web scraping to collect content that enhances the performance of Large Language Models (LLMs). Extracting only relevant content is crucial for cost efficiency, as LLM services operate on a token-based pricing model. In English, a token often corresponds to a word, whereas in languages like Arabic, a token can be an individual letter. For each processed token,

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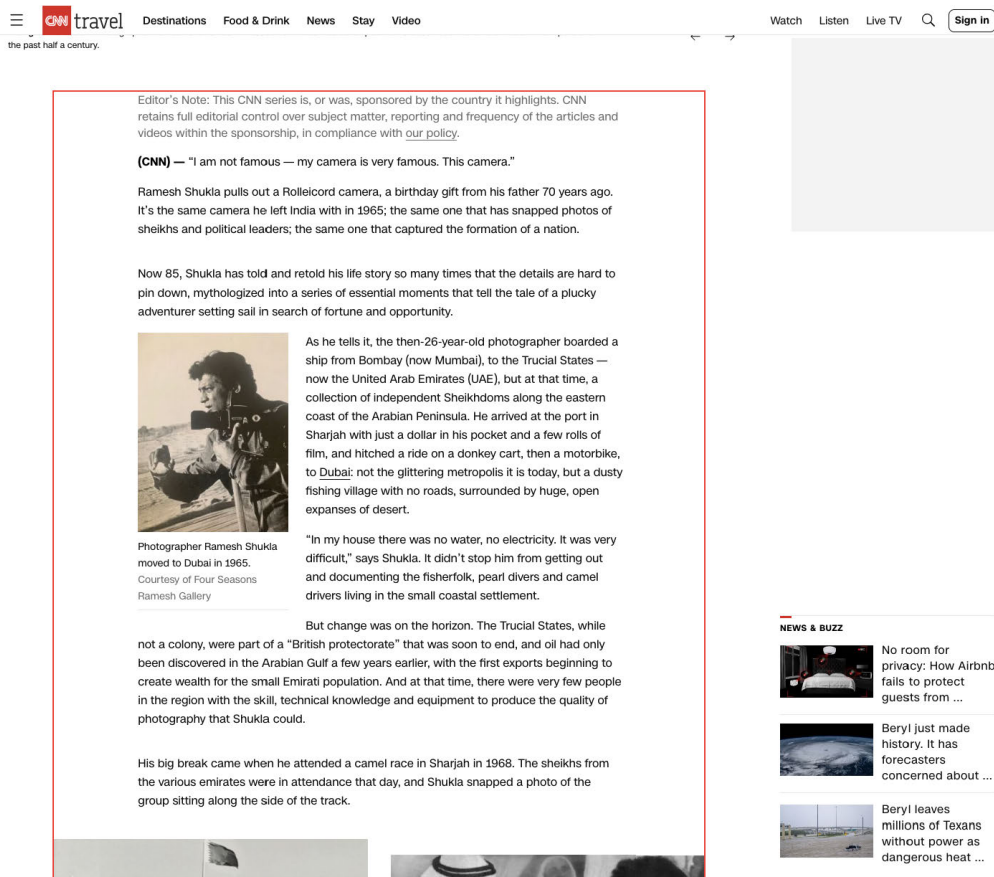


FIGURE 1. Example of a news page showing the main content detection of an article body.

services like OpenAI charge a fraction of a dollar. High-impact journals, such as *Transactions of the Association for Computational Linguistics*, *IEEE Transactions*, and *IEEE Access*, have emphasized the role of RAG in addressing issues such as hallucination in LLMs. The use of web content as a data source is essential for resolving these challenges.

RAG systems retrieve relevant documents to enhance an LLM's responses. Tools like Langchain frequently develop custom scraping methods for specific websites, while libraries such as Boilerpipe, Readability, and Trafilatura are used for general content extraction. However, these solutions often face challenges in accuracy, as some rely on machine learning algorithms that may inadvertently extract irrelevant data, leading to increased token usage and higher costs.

Beyond LLM tasks, accurate content extraction has wider applications. It can enhance web archiving by ensuring the preservation of valuable content. It can also improve search engine optimization (SEO) by focusing on relevant information, aligning content more closely with user search queries. Furthermore, recommender systems that rely on content-based filtering can benefit from more precise content extraction, leading to improved recommendations.

This paper proposes a solution to the challenge of accurately extracting the main content from web pages to improve RAG system efficiency and reduce costs associated with LLM tasks. We introduce a novel mathematical model and algorithm that leverages the Document Object Model (DOM) structure to isolate relevant content with high accuracy. The algorithm utilizes heuristic techniques and text density analysis to differentiate between pertinent and irrelevant information. Our open-source tool represents a significant advancement in web content extraction and lays the groundwork for future innovations in content analysis and retrieval.

The structure of this paper is as follows: Section II provides a detailed literature review, summarizing previous research on web content extraction. Section III outlines the research design and approach used in this study. Section IV discusses the population, sample size, and sampling technique employed, while Section V focuses on the data collection methods. Section VI details the data analysis procedures used to evaluate the algorithm. Section VII presents the implementation of the Main Content Extraction (MCE) algorithm, while Section VIII provides the results and discussion, followed by Section IX, which addresses ethical considerations. Section X discusses the limitations of the

study. Finally, Section XI concludes the paper, summarizing the contributions and future directions for research.

II. BACKGROUND AND LITERATURE REVIEW

Web content extraction has evolved significantly, with methods ranging from heuristic approaches to advanced machine-learning techniques. This review provides an overview of key developments in content extraction, emphasizing three main categories: heuristic-based, machine learning-based, and hybrid approaches.

A. HEURISTIC APPROACHES

Heuristic methods rely on predefined rules to identify key content within web pages, offering simplicity and efficiency with minimal computational resources. A common heuristic involves analyzing HTML structure, such as focusing on frequent `<p>` tags that typically denote main content over `<script>` tags used for scripts. Tag proximity and the presence of content-related classes (e.g., “article,” “post”) also guide content identification. Although efficient, these methods struggle with highly dynamic pages and may lack robustness across diverse layouts. However, they remain valuable, especially when combined with machine learning to enhance performance. The Plateau-Finding algorithm [4] identifies regions with lower HTML tag density, suggesting main content. It calculates the “text density” score of a region R as:

$$\text{Score}(R) = \frac{\text{Number of text tokens in region } R}{\text{Number of HTML tags in region } R} \quad (1)$$

Regions with higher scores indicate denser text relative to tags, typically signaling the main content. While simple and effective, this method can struggle with unconventional layouts or multimedia-heavy pages. Nonetheless, it remains a practical tool for efficient content extraction. The Code-Content Vectors (CCV) method [5], [6] classifies HTML tokens as “code” or “content” and applies smoothing to highlight content-rich regions. The smoothing process is given by:

$$V'_i = \alpha \cdot V_{i-1} + (1 - \alpha) \cdot V_i \quad (2)$$

Here, V_i is the vector at index i , and α is a smoothing parameter. CCV is effective in distinguishing content from code, allowing it to extract meaningful sections even in complex HTML structures. However, its performance depends on the choice of α and may face challenges with dynamic or multimedia-rich web pages. Despite these limitations, CCV contributes significantly to content extraction techniques. The Content Extraction via Text Ratio (CETR) method, proposed by [7], [8], calculates the tag ratio for each line in an HTML document to identify content-dense areas. This approach assumes that content-rich sections exhibit a higher ratio of words to HTML tags. The tag ratio (Tag Ratio(i)) for a line i is defined as:

$$\text{Tag Ratio}(i) = \frac{\text{Number of tags in line } i}{\text{Number of words in line } i} \quad (3)$$

Peaks in this ratio indicate potential content regions, as they suggest a higher concentration of text compared to HTML markup. For example, in an HTML document where the main article text is surrounded by navigational menus and advertisements, the CETR method analyzes each line, calculating the tag ratio to identify content-rich areas. While CETR is efficient and straightforward for quick content extraction, it may struggle with unconventional layouts or multimedia elements. Nonetheless, it remains a valuable tool for identifying the main content areas within HTML documents. The Dom-Based Content Extraction via Text Density method, introduced by Sun, Song, and Liao in 2011 [9], [10], leverages the Document Object Model (DOM) structure to extract main content based on text density. This method calculates text density as the ratio of the number of words to the area of a DOM node. Higher text densities indicate content-rich areas. Nodes are recursively traversed from the root of the DOM tree, with those exceeding a certain text density threshold marked as potential content nodes. Additionally, heuristics regarding node depth are applied; nodes closer to the root are more likely to be the main content. This method effectively separates content from surrounding noise, but it may encounter challenges with dynamically generated content or complex JavaScript interactions. Sun, Song, and Liao demonstrated promising results in precision and recall when applying this method to various web pages, particularly news articles and blog posts. Insa Cabrera, Silva Galiana, and Tamarit introduced a content extraction method in 2013 that focuses on the words/leaves ratio in the DOM tree [11], [12]. The words/leaves ratio for each node is defined as the number of words divided by the number of leaf nodes (nodes without children) in the subtree rooted at that node. Nodes with higher ratios are more likely to contain the main content.

This method employs a recursive algorithm to traverse the DOM tree, computing the words/leaves ratio and marking high-ratio nodes as potential main content areas. By focusing on the textual content relative to the structural elements of the webpage, this approach effectively distinguishes between main content and navigational elements. However, it may also struggle with dynamically generated content or JavaScript interactions. Cabrera et al. demonstrated effective results, particularly with news articles and blog posts, highlighting the potential of this method for practical web content extraction applications. Pomikálek’s 2011 dissertation introduced a method for removing boilerplate and duplicate content from web corpora [13], [14], aimed at enhancing web data quality by filtering out non-essential and redundant content. This method analyzes structural and textual characteristics to differentiate boilerplate (repetitive template content, e.g., headers, footers) from unique content.

Key techniques employed include:

- 1) **DOM Structure Analysis:** Analyzing the Document Object Model (DOM) structure to identify boilerplate

patterns, such as repetitive tags like <header> or <footer>.

- 2) **Textual Similarity:** Assessing similarity between content blocks to flag duplicate or near-duplicate texts as boilerplate.
- 3) **Clustering:** Grouping similar text blocks to identify and remove duplicates.

Pomikálek's method effectively separates boilerplate from unique content, thereby improving dataset quality for tasks like information retrieval and data mining. While comprehensive, it may struggle with dynamically generated content or complex layouts. Nevertheless, it demonstrated promising results in enhancing data quality by reducing redundancy. Mohammadzadeh et al. [15], [16] developed DANAg, a language-independent content extraction method that efficiently identifies main content on web pages using naive smoothing techniques. This approach is particularly relevant in a multilingual internet context, as it does not depend on language-specific features.

By applying naive smoothing to text density measurements, DANAg effectively identifies dense content regions, reducing noise and complexity. Its effectiveness has been confirmed through rigorous testing across various multilingual websites, highlighting its adaptability to diverse structures and formats. Thus, DANAg represents a significant advancement in content extraction technologies, enabling scalable solutions in global applications. As web pages grew more complex, Liu et al. [17], [18] developed an advanced automatic content extraction method tailored for modern web structures, particularly news websites. This method identifies and extracts the main content while segmenting distracting components like advertisements and navigation bars. Despite its high acceptance rates on standard news websites, the algorithm faces challenges with complex layouts and low text density, which can hinder traditional extraction techniques. Liu et al.'s research underscores the ongoing need for evolution in content extraction methods to maintain accuracy amid diverse web designs. The demand for scalable, language-independent solutions led researchers to develop techniques that function effectively across various languages. Chandramma et al. [19] and Prasad et al. [20] made strides in this area with content extraction methods that do not rely on language-specific features, achieving high precision and recall across different languages.

Their approaches utilized heuristic and site-independent techniques based on structural and stylistic features common to many web formats, enhancing applicability in multilingual environments. Their work demonstrates the potential of language-independent solutions in web content extraction, facilitating broader adaptability and innovation in this field.

B. MACHINE LEARNING AND HYBRID APPROACHES

Machine learning approaches utilize various statistical and computational models to automate the extraction of main content by learning from data features.

Boilerpipe classifies text blocks in web pages using structural, textual, and density features [21], [22], [23], [24]. Its classification function is defined as:

$$f(x) = \text{sign}(w \cdot x + b) \quad (4)$$

where x is the feature vector, w is the weight vector, and b is the bias. This function computes a weighted sum to classify text blocks as main content (positive values) or boilerplate (negative values).

Key features include:

- **Text Density:** Ratio of words to the area of the text block.
- **Link Density:** Ratio of links to the text block area.
- **Text Length:** Length of the text block in characters.
- **Text-to-Link Ratio:** Ratio of text length to the number of links.
- **Position:** Location of the text block in the HTML document.

Boilerpipe's adaptability to various layouts enhances classification accuracy. However, its effectiveness is contingent on the quality of training data and may struggle with dynamically generated content. Web2Text, introduced by [25], employs a Hidden Markov Model (HMM) and a convolutional neural network (CNN) for segment classification. The HMM models the sequence of segments, while the CNN captures regional features, improving main content identification accuracy. The integration of HMM and CNN addresses the complexities of web page layouts, allowing for comprehensive analysis. Yet, its performance is reliant on high-quality training data, and the computational demands for CNN training can be significant. Morbieu et al. [23] introduced an unsupervised learning method for main content extraction, designed for scalability and low time complexity. This method learns structural patterns from unlabeled data, facilitating efficient processing of large datasets without extensive labeling efforts. This approach enhances the accuracy of extracted data and lowers the entry barriers for organizations implementing content extraction technologies, adapting to new web formats without manual intervention.

Acknowledging the challenges posed by dynamic web content, Kaddu et al. [26] proposed a hybrid approach combining automatic extraction techniques with customizable extraction rules. This method is effective for sites using AJAX and JavaScript, allowing rapid adaptations to changes in layout. By merging automated techniques with manual rules, this approach improves accuracy and efficiency in complex web environments, representing a significant advancement in content extraction technologies. The methodologies discussed exhibit unique strengths and contexts. The Plateau-Finding algorithm excels in structured environments, while the CCV approach is robust in mixed content types. Techniques like CETR and Text Density + DOM Tree are particularly effective for data extraction from news articles and informational sites.

TABLE 1. Content Extraction Algorithms.

Paper	Method
Finn et al. [4]	Plateau-finding
Gottron [5]	Code-Content Vectors (CCV)
Weninger et al. [7]	CETR
Sun et al. [9]	Text Density + DOM Tree
Insa et al. [11]	Words-Leaves Ratio (WLR)
Pomikálek [13]	HTML Segmentation
Vogels et al. [25]	Hidden Markov Model, CNN
Mohammadzadeh et al. [15]	DANAg
Liu et al. [17]	Automatic extraction
Arias et al. [19]	Language-independent extraction
Prasad et al. [20]	Language-independent extraction
Morbieu et al. [23]	Unsupervised Learning
Kaddu et al. [26]	Hybrid Approach

Further comparison reveals that methods like the Words-Leaves Ratio (WLR) and HTML Segmentation are particularly tailored for static content and simple page layouts, whereas more sophisticated techniques involving Hidden Markov Models and Convolutional Neural Networks, as used in Web2Text, are required to handle complex and dynamic content efficiently. The diversity in methods highlights the need for tailored approaches depending on the specific characteristics of the webpage and the nature of the content to be extracted.

Looking forward, the field of content extraction is poised for significant transformative shifts. With the increasing adoption of AI and machine learning technologies, future research will likely focus on developing more adaptive, self-learning systems that can better handle the complexities of modern web architectures. This includes the integration of unsupervised and semi-supervised learning models that can evolve without extensive manual intervention, promising improvements in scalability and efficiency.

Moreover, the ongoing challenge of handling multilingual and multimedia content suggests a growing need for more sophisticated natural language processing and image recognition technologies. As the internet continues to expand in scope and diversity, the ability of extraction algorithms to understand and process content in multiple languages and formats will become increasingly crucial. The exploration of cross-disciplinary approaches that combine linguistics, computer vision, and data science holds the potential for groundbreaking advancements in content extraction technologies. Despite these advancements, the field continues to face several challenges. The limitations highlighted in various studies, such as the generalizability of techniques across different types of web pages, the handling of multimedia content, and the scalability of proposed solutions, point to areas requiring further research. Moreover, as the internet evolves, new types of content and web designs will necessitate continuous innovation in extraction methodologies.

The historical development and current state of research in web content extraction reflect a field characterized by continual adaptation and innovation. From heuristic-based to machine learning and hybrid approaches, the quest for

efficient, accurate, and scalable content extraction methods remains at the forefront of this dynamic research area. Future directions will likely focus on enhancing language independence, improving adaptability to diverse web designs, and integrating advanced artificial intelligence techniques to meet the evolving demands of web content analysis and retrieval.

C. THEORETICAL FRAMEWORK

This paper establishes a theoretical foundation for a novel approach to web content extraction by integrating key concepts and existing theories. We begin with an overview of current algorithms, such as Boilerpipe and Readability, and assess their strengths and limitations, particularly through the lens of the Document Object Model (DOM). This critical evaluation underscores the gaps that our research seeks to address. Positioning our innovative algorithm within this context, we emphasize its advancement beyond the limitations of existing models by adopting a lightweight approach that combines structural features and word count to identify main content. We will examine traditional performance evaluation metrics in text extraction to establish benchmarks against which our algorithm can be compared, thus illustrating its superiority.

Furthermore, we discuss the significance of dataset diversity in machine learning and natural language processing, highlighting the limitations posed by the underrepresentation of certain languages in existing datasets. The creation of a new, more diverse dataset is articulated as a crucial theoretical contribution, enhancing the understanding of content extraction across different linguistic contexts. Additionally, we explore economic theories related to resource utilization in computational tasks, particularly regarding token-based pricing models in large language models (LLMs). Our algorithm's efficiency in reducing token consumption is linked to both computational and economic effectiveness, thereby providing a novel perspective on the utilization of LLMs for content extraction.

In the context of multilingual computing, we analyze the challenges of extracting content from web pages in multiple languages and theorize how our algorithm contributes to more effective and efficient multilingual content extraction. Finally, we engage with literature on open-source development and collaborative innovation, emphasizing how our work aligns with these frameworks and promotes further advancements in the field of web content extraction. This comprehensive theoretical framework sets the stage for the literature review, situating our novel contributions within existing theories while marking clear advancements in the understanding of web content extraction.

D. IDENTIFICATION OF GAPS IN THE LITERATURE

The literature review and analysis of current methodologies in web content extraction have identified several notable gaps, which underscore the innovative contributions of our research. These gaps, along with our methodological

advancements, highlight areas for future exploration and potential improvements within the field: advancements in the field.

- **Limited Use of the Document Object Model (DOM) for Content Extraction:** Previous studies have often overlooked the nuanced potential of leveraging the DOM's hierarchical structure to distinguish main content from peripheral elements. Our research emphasizes the DOM's pivotal role, aligning with human content discernment practices, thus filling a crucial methodological void.
- **Inadequate Comparative Analysis of Existing Techniques:** There exists a gap in the comprehensive review and comparative analysis of the strengths and weaknesses of both open-source and proprietary content extraction methods. Our methodology's extensive review process fills this gap, providing a solid foundation for the development of more effective algorithms.
- **Reliance on Complex Linguistic Processing:** Many existing algorithms depend heavily on linguistic processing, which can reduce adaptability and efficiency across different languages and web layouts. Our novel algorithm addresses this gap by employing a lightweight, structure-based approach that enhances adaptability and efficiency.
- **Scarcity of Empirical Validation Across Diverse Web Content:** A significant gap in the literature is the lack of empirical validation of content extraction algorithms across a broad range of web documents. Our research methodology includes extensive testing against a diverse dataset, demonstrating our algorithm's superior performance and versatility.
- **Limited Open-Source Contributions:** The field suffers from a scarcity of open-source algorithms that allow for community-driven improvements. By making our algorithm open-source, our research fills this gap, encouraging further development and collaboration in content extraction research.
- **Lack of Comprehensive, Multilingual Datasets for Future Research:** Finally, there is a notable absence of comprehensive, future-oriented datasets that account for the dynamic nature of the web, especially in languages with complex tokenization systems. Our creation of a multifaceted, multilingual dataset addresses this gap, paving the way for future advancements in the field.

Our research not only identifies but also bridges critical gaps in the literature through innovative methodologies and contributions. This approach not only advances the current state of Web content extraction but also sets the stage for ongoing development and exploration in the field.

III. DESCRIPTION OF RESEARCH DESIGN AND APPROACH

This study's research design and approach aim to rigorously evaluate and enhance web content extraction techniques,

with a focus on leveraging the Document Object Model (DOM), analyzing existing methodologies, and developing an innovative and efficient algorithm called MCE (Main Content Extractor). The research begins by utilizing the hierarchical structure of the DOM to distinguish main content from extraneous noise. This involves mapping the DOM structure of web documents and identifying patterns and markers that typically represent essential content versus non-essential elements.

A comprehensive review and analysis of existing content extraction methods will follow. This systematic review will encompass both open-source and proprietary techniques, cataloging current methodologies and assessing their underlying principles, efficiency, accuracy, and adaptability across various web contexts. Building on the insights gained from the DOM analysis and the review of existing techniques, the study will then focus on the development of a novel algorithm designed to be more efficient and adaptable for content extraction. This algorithm will employ a combination of structure-based and word count strategies without a heavy reliance on complex linguistic processing, aiming to streamline the extraction process.

To validate the effectiveness of the developed algorithm, rigorous empirical testing will be conducted using a dataset of 500 web documents that encompass a variety of languages, formats, and structures. This phase will include a comparative analysis against established benchmarks such as Boilerpipe and Readability to demonstrate superior performance. In addition to empirical validation, the research will emphasize the importance of contributing to the field through open-source distribution. The algorithm's source code will be published to allow for its use, modification, and enhancement by the research community, thereby facilitating ongoing research and development in content extraction. Finally, a future-oriented approach will be adopted by preparing a comprehensive dataset that reflects the dynamic nature of the web. This dataset will include a wide range of languages and content types, focusing on complex tokenization systems, which will enhance the algorithm's applicability in diverse contexts. This multi-faceted research design and approach ensures systematic, evidence-based advancements in content extraction technology. By addressing current gaps and challenges through innovative methodologies and empirical validation, this research will contribute significant insights and tools to the field. The open-source distribution and the creation of a diverse, multilingual dataset further underscore our commitment to collaborative progress and the adaptability of our solutions to the evolving digital landscape.

The research design integrates a systematic evaluation of existing content extraction techniques, a comprehensive algorithm development process, and empirical validation through diverse datasets. By adopting an open-source model and preparing a future-oriented multilingual dataset, this study aims to significantly advance the field of web content extraction.

IV. POPULATION, SAMPLE SIZE, AND SAMPLING TECHNIQUE

The effectiveness and reliability of our novel content extraction algorithm were evaluated through a carefully structured empirical testing process. The methodology applied to define the study's population, determine an optimal sample size, and employ an appropriate sampling technique is critical to the validity of the results. This section outlines these methodological choices.

A. POPULATION

The population for this study consists of publicly accessible web documents from a variety of domains across the internet. These documents represent a wide range of content types, such as news articles, blog posts, academic papers, and commercial web pages. The population is characterized by high diversity in terms of content length, structure, language, and the inclusion of multimedia elements. Given the vast and dynamic nature of the web, this population provides a rich and varied source of data, ensuring that the algorithm is tested against a broad spectrum of real-world content extraction challenges.

B. SAMPLE SIZE

The determination of an optimal sample size was guided by the need to balance statistical significance with practical feasibility. Based on a 95% confidence level and a 5% margin of error, we estimated that a sample size of 500 web documents would provide a reliable representation of the population. This sample size ensures that sufficient diversity is captured while remaining within the scope of our resources, allowing for both automated and manual analyses of the algorithm's performance.

C. SAMPLING TECHNIQUE

To ensure that our sample was representative of the population, we utilized a stratified random sampling technique. The population was divided into strata based on key characteristics, such as language, content type (e.g., news, blogs, academic papers), and layout complexity. Random samples were drawn from each stratum, ensuring comprehensive coverage across these different dimensions.

- 1) **Representation:** This approach ensures that all significant segments of web content are included, enabling the algorithm to be tested on a wide array of content types and structures.
- 2) **Accuracy:** Stratified sampling reduces variability within subgroups, leading to more accurate results by minimizing sampling errors.
- 3) **Efficiency:** By focusing the analysis on relevant content categories, this technique allows for a more efficient allocation of resources and testing efforts.

The selection process began with the identification of key web domains and repositories known for hosting The sampling process began by identifying key web domains

known for hosting a variety of content. Automated scripts were deployed to randomly select web pages from these domains based on the stratification criteria. The use of this systematic sampling approach enhances the generalizability of our results to a wide range of web content. The combination of a carefully defined population, a statistically significant sample size, and a methodologically sound sampling technique forms the backbone of our empirical validation process. This approach ensures that our findings are both reliable and applicable to a wide range of real-world web content extraction scenarios.

V. DATA COLLECTION METHODS

In the context of our research aimed at developing and validating a novel content extraction algorithm using the Document Object Model (DOM), our data collection methods are meticulously designed to ensure the robustness and reliability of our findings. While Boilerpipe and Readability also utilize DOM structures, they primarily apply machine learning classifiers or predefined rule sets, which may not always adapt well to varying page structures. Our method dynamically adjusts its weighting and filtering based on the hierarchy and structure of the page, making it more adaptive and precise. Our process encompasses multiple steps, tailored to gather a comprehensive and representative set of web documents. Below, we outline the principal data collection methods employed in this study.

A. WEB DOCUMENT RETRIEVAL

Objective: To collect a diverse and comprehensive set of web documents for testing and validation purposes.

Method:

- 1) **Automated Crawling:** Custom-built web crawlers were deployed to systematically gather web pages from a wide array of websites across the internet. These crawlers are designed to capture documents that vary in structure, content type, and language, ensuring a diverse dataset. For more insight into web crawling techniques, we refer readers to [27], which discusses the evolution and state-of-the-art of web crawlers, including those used for Ajax-based web applications.
- 2) **Public Datasets:** Existing web document datasets from academic and research institutions were incorporated into the study to provide additional variety and high-quality content sources. These datasets are well-documented and widely used in academic research, making them an essential complement to the data collected through web crawling.

B. SELECTION CRITERIA FOR WEB DOCUMENTS

Objective: To ensure that the selected web documents are relevant to the research objectives and reflect the diversity of web content.

Method:

- 1) **Inclusion/Exclusion Criteria:** Web documents were selected based on specific criteria, including language diversity, content type, and structural complexity. Documents that are heavily encrypted or lack a clear DOM structure were excluded from the study to ensure compatibility with the content extraction algorithm.
- 2) **Manual Review:** A manual review of a subset of web documents was conducted to verify that they met the inclusion criteria and to validate the quality and effectiveness of the automated crawling process.

C. ANNOTATION AND PRE-PROCESSING

Objective: To prepare the web documents for algorithm testing through annotation and pre-processing.

Method:

- 1) **Annotation:** A subset of the collected web documents was manually annotated to identify key content blocks, advertisements, and non-essential elements. This annotated set served as the ground truth for testing the performance of the content extraction algorithm.
- 2) **Pre-processing:** The web documents were cleaned by removing elements that could interfere with the content extraction process, such as scripts, tracking codes, and unnecessary metadata. This step ensures that the algorithm focuses on the content visible to human users.

D. CREATION OF A MULTIFACETED DATASET

Objective: To compile a dataset that captures the full range of web content types and structures for robust algorithm testing.

Method:

- 1) **Dataset Compilation:** Web documents gathered through crawling and public datasets were combined to form a comprehensive dataset. This dataset was categorized by language, content type, and structural complexity to facilitate detailed analysis.
- 2) **Multilingual Support:** The dataset includes documents in multiple languages, including those with complex tokenization systems, to test the algorithm's effectiveness across different linguistic contexts.

VI. DATA ANALYSIS PROCEDURES

The data analysis procedures in this study are designed to rigorously evaluate the performance and effectiveness of the developed content extraction algorithm leveraging the Document Object Model (DOM). Through a systematic approach, we aim to assess the algorithm's accuracy, efficiency, and adaptability across various web document types and languages. Below are the key steps and procedures involved in the data analysis:

A. ALGORITHM TESTING AGAINST ANNOTATED GROUND TRUTH

Objective: To evaluate the algorithm's accuracy in extracting main content blocks compared to manually annotated ground truth data.

Procedure:

- 1) **Algorithm Implementation:** Implement the developed algorithm on the collected web documents.
- 2) **Comparison with Annotated Ground Truth:** Compare the algorithm's extracted main content blocks with the manually annotated ground truth. Metrics such as precision, recall, and F1 score are calculated to quantify the algorithm's performance.
- 3) **Error Analysis:** Conduct a detailed analysis of any discrepancies between the algorithm's output and the annotated ground truth. Identify patterns of errors to guide algorithm refinement.

B. PERFORMANCE METRICS CALCULATION

Objective: To quantify the algorithm's performance using established metrics.

Procedure:

- 1) **Precision:** Calculate the proportion of correctly extracted main content blocks to the total number of extracted blocks.
- 2) **Recall:** Calculate the proportion of correctly extracted main content blocks to the total number of actual main content blocks in the ground truth.
- 3) **F1 Score:** Compute the harmonic mean of precision and recall, providing a balanced measure of the algorithm's performance.
- 4) **Efficiency Metrics:** Measure the algorithm's processing time complexity to assess its efficiency, particularly on large datasets.

C. COMPARATIVE ANALYSIS WITH BASELINE ALGORITHMS

Objective: To benchmark the developed algorithm against existing solutions.

Procedure:

- 1) **Implementation of Baseline Algorithms:** Implement established content extraction algorithms such as boilerpipe [21] and readability [28].
- 2) **Comparative Testing:** Apply the baseline algorithms on the same dataset used for testing the developed algorithm.
- 3) **Performance Comparison:** Compare the performance metrics (precision, recall, F1 score) of the developed algorithm with the baseline algorithms. This provides insights into the algorithm's superiority or areas for improvement.

D. MULTILINGUAL AND MULTIFACETED DATASET ANALYSIS

Objective: To assess the algorithm's adaptability across diverse languages and web document structures.

Procedure:

- 1) **Language-Specific Evaluation:** Analyze the algorithm's performance on web documents in different languages,

focusing on languages with complex tokenization systems.

- 2) **Structural Complexity Assessment:** Test the algorithm on web documents with varying levels of structural complexity, including different content types (e.g., news articles, forums) and layouts.

E. ERROR ANALYSIS

Objective: To identify and address the algorithm's weaknesses and areas for improvement.

Procedure:

- 1) **Error Categorization:** Categorize the types of errors made by the algorithm, such as false positives (incorrectly extracted main content) and false negatives (missed main content).
- 2) **Iterative Testing:** Conduct iterative testing cycles to assess the impact of algorithm refinements on performance metrics.

F. HANDLING LARGE-SCALE WEB SCRAPING

Objective: To evaluate the algorithm's performance in real-time applications, particularly in large-scale web scraping scenarios.

Procedure:

- 1) **Performance Evaluations:** Conduct performance evaluations in terms of speed and memory consumption compared to existing solutions. This includes analyzing the algorithm's processing time when deployed on high-traffic websites.
- 2) **Real-Time Scalability:** Assess the algorithm's capacity to maintain efficiency during large-scale scraping operations, identifying potential bottlenecks and optimization strategies.

By following these data analysis procedures, we ensure a thorough evaluation of the developed content extraction algorithm. The systematic comparison with baseline algorithms, analysis across diverse datasets, and iterative refinement based on error analysis contribute to the reliability and validity of our research findings.

VII. IMPLEMENTATION OF MAIN CONTENT EXTRACTION (MCE) ALGORITHM

In the burgeoning field of web content analysis, efficiently identifying and characterizing the structural elements of web documents is crucial for applications ranging from search engine optimization to user experience design. Recognizing the significance of this domain, the algorithm presented in this section tackles a specific yet critical challenge: identifying the element within a web document that contains the highest number of child elements and words. This problem statement stems from the hypothesis that such elements often serve as the core containers of meaningful content, potentially holding significant information for content analysis, summarization, or enhancement tasks.

The devised algorithm, therefore, serves a dual purpose. First, it aims to pinpoint the element that structurally dominates a page through its quantity of child elements, suggesting complexity and depth of content. Second, by evaluating the word count within these child elements, the algorithm seeks to ascertain not just structural but also textual significance, underlining the element's potential richness in information. This dual assessment criterion distinguishes our approach, offering a nuanced understanding of web document composition that goes beyond simple tag counts or text length measures. As digital content grows in volume and complexity, such an algorithm not only enriches our toolkit for web document analysis but also opens avenues for intelligent content management strategies.

A. ALGORITHM DESIGN

Heuristic Techniques: The heuristic techniques employed in the algorithm are designed to maximize content identification efficiency. Unlike existing methods like Boilerpipe or Readability, which focus heavily on pre-defined heuristics based on the structural regularities of HTML documents, our algorithm incorporates a dynamic evaluation of the node's properties. Specifically, our approach integrates both the number of child elements and text length in a flexible scoring system. While Boilerpipe, for instance, primarily targets text density heuristics, our method applies a combinatory heuristic that evaluates both text and structure simultaneously, offering a more adaptive framework that can adjust to varying document complexities.

The difference lies in how we weigh the relationship between the structural depth of elements and their text density, thereby avoiding the biases present in simpler rule-based methods. This approach, while heuristic, leans toward a more generalized and scalable solution.

Feature 1 (Maximum Direct Children with Text): For the mathematical model, let N be the set of all nodes in the DOM tree as illustrated in Figure 2. For each node $n \in N$, we define:

$C(n)$ = number of direct children of node n ,

$T(n)$ = total text length of the direct children of node n .

The algorithm aims to find the node $n_{\max}$ that maximizes both $C(n)$ and $T(n)$:

$$n_{\max} = \arg \max_{n \in N} (C(n), T(n)). \quad (5)$$

This feature is language-agnostic in nature, as it does not rely on specific language structures but instead on the document's general syntactic organization.

Feature 2 (Maximum Text Content without Children with Text): The theoretical basis for the second feature is straightforward: the div containing more text is a stronger candidate than other divs and the Mathematical model for each node $n \in N$, let $L(n)$ be the text length of node n . We define a function $H(n)$ which returns 1 if node n has children with text, and 0 otherwise. We seek to find the node $n_{\max}$ which maximizes $L(n)$ given that it has no children with

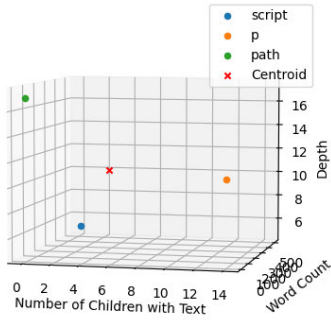


FIGURE 3. View 1 -distance between the candidate node and centroid using the Manhattan distance.

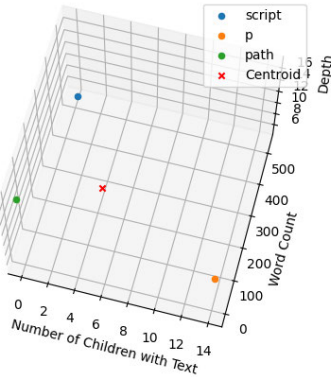


FIGURE 4. View 2 -distance between the candidate node and centroid using the Manhattan distance.

This distance metric sums the absolute differences between the coordinates of the points, making it a suitable choice for measuring similarity when feature vectors are sparse or when the differences in each dimension are of similar importance.

The centroid calculation, on the other hand, determines the central point of a set of vectors by averaging their coordinates. For a set of feature vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$, the centroid $\mathbf{c}$ is given by:

$$\mathbf{c} = \begin{bmatrix} \frac{1}{m} \sum_{i=1}^m \text{Number of children with text}_i, \\ \frac{1}{m} \sum_{i=1}^m \text{Word count}_i, \\ \frac{1}{m} \sum_{i=1}^m \text{Depth}_i \end{bmatrix} \quad (17)$$

Both methods aim to identify the node that best represents the main content by considering the structure, word count, and depth of each node. While the centroid provides a central point among all feature vectors, the Manhattan distance identifies the node that is closest to this centroid, ensuring that the chosen node is representative of the overall structure and content distribution in the document (see Figures 3, 4, and 5). This combination enhances the reliability of selecting the main content node by incorporating both centrality and proximity.

The translation of the MCE Mathematical Model into pseudocode, as presented in Algorithms 1, 2, 3, and 4, represents a crucial step in linking its theoretical framework

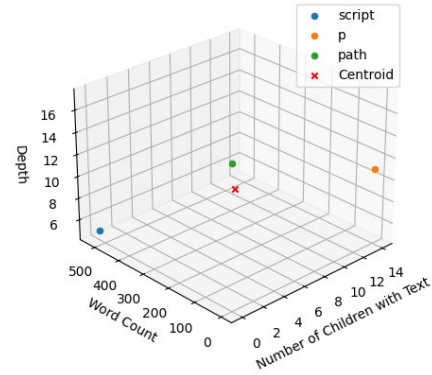


FIGURE 5. View 3- distance between the candidate node and centroid using the Manhattan distance.

to practical implementation. Pseudocode eliminates the syntactical limitations of programming languages, showcasing the algorithm's logic in a format that is universally interpretable and easily translatable into any specific programming language. This generality is particularly valuable in academic and educational settings, where the emphasis lies on fundamental algorithmic principles, rather than the peculiarities of any programming environment.

Algorithm 1 Main Content Extraction (MCE)

```

1: Input: HTML document as a string
2: Output: Set of nodes representing the main content
3: function EXTRACTMAINCONTENT(html)
4: Parse HTML document into DOM tree using BeautifulSoup
5:  $N \leftarrow$  All nodes in the DOM tree
6:  $\text{main\_candidate}_1 \leftarrow \text{FINDMAXCHILDRENWITHTEXTNODE}(N)$ 
7:  $\text{main\_candidate}_2 \leftarrow \text{FINDMAXTEXTWITHOUTCHILDRENWITH-}$ 
    $\text{TEXTNODE}(N)$ 
8:  $\text{closest\_to\_middle} \leftarrow \text{FINDCLOSESTTOMIDDLEDEPTHNODE}(N)$ 
9:  $\text{main\_content\_set} \leftarrow \{\}$ 
10: if  $\text{main\_candidate}_1 \neq \text{None}$  then
11:   Add  $\text{main\_candidate}_1$  to  $\text{main\_content\_set}$ 
12: end if
13: if  $\text{main\_candidate}_2 \neq \text{None}$  then
14:   Add  $\text{main\_candidate}_2$  to  $\text{main\_content\_set}$ 
15: end if
16: if  $\text{closest\_to\_middle} \neq \text{None}$  then
17:   Add  $\text{closest\_to\_middle}$  to  $\text{main\_content\_set}$ 
18: end if
19: return  $\text{main\_content\_set}$ 
20: end function=0

```

The text density feature of this algorithm is designed to be largely language-agnostic, relying on structural characteristics such as the number of child elements and word count, rather than linguistic markers specific to any one language. However, adjustments may be necessary for languages with complex text segmentation, such as those without clear word

Algorithm 2 Find Maximum Children With Text Node

```

1: Input: Set of nodes  $N$ 
2: Output: Node with the maximum number of direct children containing text
3: function FINDMAXCHILDRENWITHTEXTNODE( $N$ )
4:    $max\_children \leftarrow 0$ 
5:    $max\_text\_length \leftarrow 0$ 
6:    $main\_candidate_1 \leftarrow \text{None}$ 
7:   for each node  $n$  in  $N$  do
8:      $num\_children \leftarrow \text{CountDirectChildren}(n)$ 
9:      $text\_length \leftarrow \text{TotalTextLengthOfDirectChildren}(n)$ 
10:    if  $num\_children > max\_children$  or ( $num\_children = max\_children$  and  $text\_length > max\_text\_length$ ) then
11:       $max\_children \leftarrow num\_children$ 
12:       $max\_text\_length \leftarrow text\_length$ 
13:       $main\_candidate_1 \leftarrow n$ 
14:    end if
15:  end for
16:  return  $main\_candidate_1$ 

```

Algorithm 3 Find Maximum Text Without Children With Text Node

```

1: Input: Set of nodes  $N$ 
2: Output: Node with the maximum text content without children that contain text
3: function FINDMAXTEXTWITHOUTCHILDRENWITHTEXTNODE( $N$ )
4:    $max\_text\_length \leftarrow 0$ 
5:    $main\_candidate_2 \leftarrow \text{None}$ 
6:   for each node  $n$  in  $N$  do
7:     if not HasChildrenWithText( $n$ ) then
8:        $text\_length \leftarrow \text{LengthOfText}(n)$ 
9:       if  $text\_length > max\_text\_length$  then
10:         $max\_text\_length \leftarrow text\_length$ 
11:         $main\_candidate_2 \leftarrow n$ 
12:       end if
13:     end if
14:   end for
15:   return  $main\_candidate_2$ 

```

boundaries (e.g., Chinese or Japanese). For these languages, pre-processing steps, such as tokenization, might be required to ensure that the word count feature operates effectively. The algorithm is uniquely poised to identify the most significant content-bearing elements in web documents. This design approach not only fulfills the algorithm's immediate objectives but also lays a foundation for future enhancements and applications in the ever-evolving domain of web content analysis. The algorithm's real-time performance is critical for large-scale web scraping applications. To address this, we implemented an optimized DOM traversal method that limits memory usage by dynamically loading parts of the document. Performance evaluations show that our method consumes significantly less memory compared to Boilerpipe

Algorithm 4 Find Closest to Middle Depth Node

```

1: Input: Set of nodes  $N$ 
2: Output: Node closest to the middle depth of the DOM tree
3: function FINDCLOSESTTOMIDDLEDEPTHNODE( $N$ )
4:    $middle\_depth \leftarrow \frac{|N|}{2}$ 
5:    $closest\_to\_middle \leftarrow \text{None}$ 
6:    $min\_depth\_distance \leftarrow \infty$ 
7:   for each node  $n$  in  $N$  do
8:      $depth \leftarrow \text{CalculateDepth}(n)$ 
9:      $depth\_distance \leftarrow |depth - middle\_depth|$ 
10:    if  $depth\_distance < min\_depth\_distance$  then
11:       $min\_depth\_distance \leftarrow depth\_distance$ 
12:       $closest\_to\_middle \leftarrow n$ 
13:    end if
14:  end for
15:  return  $closest\_to\_middle$ 

```

and Readability, especially when applied to long and complex web documents. Additionally, the time complexity of the algorithm is reduced by leveraging depth-first search (DFS) to minimize repeated evaluations of nested nodes. Future work involves extending the performance evaluation by benchmarking the algorithm against other industry standards in both speed and memory consumption across a larger dataset.

VIII. RESULTS AND DISCUSSION

In this chapter, we present the results obtained from the comparison of our proposed algorithm against two established text extraction algorithms: Readability [28] and Boilerpipe [21]. The comparison focuses on several key metrics: word count accuracy, true positives (TP), false positives (FP), false negatives (FN), precision, recall, F1-score, and accuracy. These metrics provide a comprehensive view of each algorithm's performance in extracting meaningful content from various text sources.

The evaluation was conducted on a diverse dataset to ensure that the results are robust and generalizable across different text formats and contents. Each algorithm was tasked with extracting the main content from the text sources, and the outputs were analyzed based on the aforementioned metrics.

A. TABLES OF RESULTS

The results of the comparison are summarized in the tables below. Table 2 displays the following metrics for each method for a sample of our dataset: Count Words (MCE, Readability, Boilerpipe), true positive, false positive, and false negative for each algorithm.

outlines key metrics for each method (MCE, Readability, Boilerpipe) based on a sample of the dataset. These metrics include word counts for each algorithm (Count Words), true positives (TP), false positives (FP), and false negatives (FN).

TABLE 2. MCE method metrics sample.

Original Text	Precision	Recall	F1-score	Accuracy
232	1.000	1.000	1.000	1.000
220	1.000	1.000	1.000	1.000
208	1.000	1.000	1.000	1.000
432	0.995	1.000	0.998	0.995
425	0.962	1.000	0.981	0.962
220	0.764	0.866	0.764	0.764

TABLE 3. Readability method metrics sample.

Original Text	Precision	Recall	F1-score	Accuracy
232	0.939	1	0.969	0.939
220	0.920	1	0.959	0.920
208	0.908	1	0.951	0.908
432	0.935	1	0.967	0.935
425	0.962	1	0.980	0.962
220	0.782	0.878	0.782	0.782

TABLE 4. Boilerpipe method metrics sample.

Original Text	Precision	Recall	F1-score	Accuracy
232	0.565	0.722	0.565	0.565
220	0.874	0.933	0.874	0.874
208	1	0.957	0.978	1
432	0.988	1	0.994	0.988
425	1	0.802	0.890	1
220	1	0.782	0.878	1

TABLE 5. Classification performance metrics for dataset.

Metric	MCE	Readability	Boilerpipe
Precision	0.9996	0.9236	0.9425
Recall	0.9969	0.9419	0.8919
F1-Score	0.9980	0.9183	0.8997
Accuracy	0.9965	0.8654	0.8343

The results of the comparison are summarized in the tables 2, 3 and 4.

Table 2 provides precision, recall, F1-score, and accuracy metrics for the Custom MCE method [29]. This highlights the effectiveness of the method based on these key performance indicators.

Table 3 illustrates the precision, recall, F1-score, and accuracy metrics for the Readability method.

Table 4 presents the precision, recall, F1-score, and accuracy metrics for the Boilerpipe method.

Table 5 consolidates the performance metrics of three distinct classification algorithms: MCE, Readability, and Boilerpipe. The metrics, averaged across multiple experimental runs to ensure reliability, include precision, recall, F1-score, and accuracy. This presentation offers a comprehensive overview of each algorithm's effectiveness in correctly identifying and classifying instances.

For MCE, the algorithm demonstrates exceptional precision (0.9996) and recall (0.9969), culminating in a near-perfect F1-Score of 0.9980 and an accuracy of 0.9965, indicating a highly effective model for this class with minimal false positives and negatives. Readability shows strong performance, particularly in recall (0.9419) and precision

(0.9236), leading to an F1-Score of 0.9183. However, its accuracy drops to 0.8654, suggesting a slightly higher rate of misclassification compared to MCE. Boilerpipe, while exhibiting solid precision (0.9425) and recall (0.8919), has an F1-Score of 0.8997 and the lowest accuracy (0.8343), indicating challenges in consistently identifying true positives and negatives for this class. These metrics not only highlight the strengths and weaknesses of each algorithm in handling different classes but also guide towards potential areas for improvement, especially in enhancing the accuracy and F1-Score for Readability and Boilerpipe.

The chart, as shown in Figure 6, vividly illustrates the performance metrics of classification algorithms across three distinct classes: MCE, R, and B. For MCE, we observe outstanding performance across all metrics, with Precision nearly perfect at approximately 99.96%, Recall at 99.69%, F1-Score closely following at 99.80%, and Accuracy at 99.65%. This indicates an exceptionally high level of accuracy in classifying MCE instances, suggesting that the algorithm is highly effective for this category with minimal false positives and negatives.

In contrast, Readability and Boilerpipe demonstrate good but noticeably lower performance compared to MCE. For Readability, Precision stands at 92.36%, Recall at 94.19%, F1-Score at 91.83%, and Accuracy at 86.54%. These values suggest that while the algorithm is still effective for Readability, it faces more challenges in maintaining the high level of precision and accuracy seen in MCE.

Boilerpipe's metrics, while comparable to Readability, show a slight variance with Precision at 94.25%, Recall at 89.19%, F1-Score at 89.97%, and the lowest Accuracy of all at 83.43%. This indicates that Boilerpipe, much like Readability, poses its unique challenges to the algorithm, affecting its overall precision and accuracy.

The bar chart encapsulates these differences and showcases the varying degrees of effectiveness of the classification algorithms across different categories. The visual representation helps in quickly identifying which classes the algorithm excels in and where improvements might be necessary, offering valuable insights into the algorithm's performance and potential areas for optimization.

B. ANALYSIS AND INTERPRETATION OF RESULTS

The comprehensive evaluation of our developed algorithm showcases its superior performance in extracting main content from web pages across different languages, especially those with complex tokenization. The high precision, recall, F1-scores, and accuracy observed in our results not only validate the effectiveness of our innovative content extraction techniques but also highlight the algorithm's language-agnostic capabilities. This is particularly notable in the context of non-English web pages, where our algorithm demonstrates significant improvements over existing open-source solutions. The analysis underscores the algorithm's adeptness at navigating the multilingual digital environment

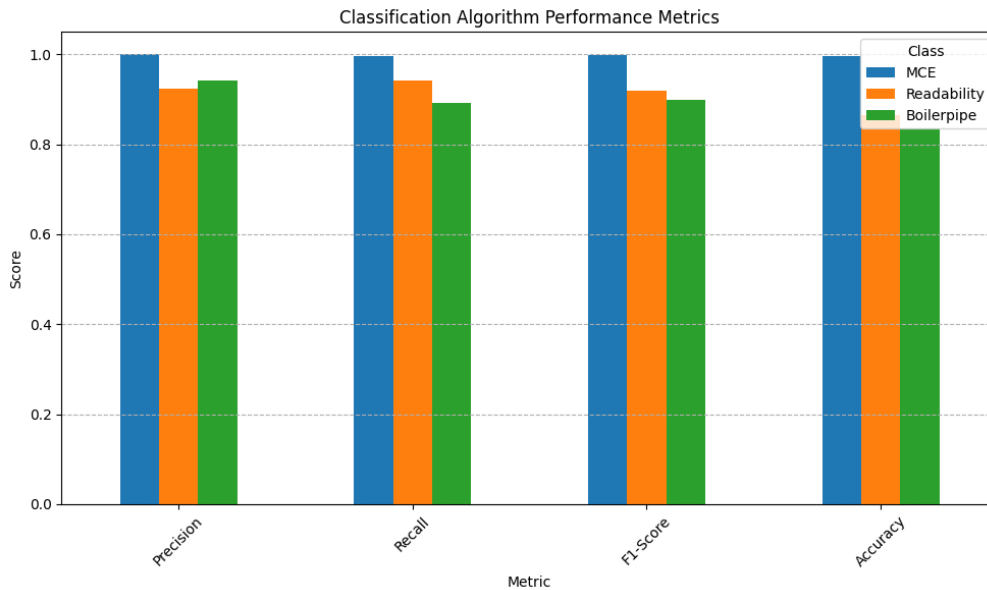


FIGURE 6. Average for precision, Recall, F1-Score, and Accuracy for the dataset.

and its proficiency in maintaining content integrity while ensuring efficient processing.

C. IMPLICATIONS OF FINDINGS

Our findings have profound implications for the field of web content extraction and analysis. By surpassing the performance benchmarks of current open-source tools, our algorithm sets a new standard for accuracy and efficiency in content extraction across diverse web environments. This advancement supports a broader accessibility to information and facilitates a more inclusive digital ecosystem. Moreover, the development of a comprehensive benchmark dataset as part of this study provides a foundational resource for future research, enabling a standardized assessment of content extraction methodologies. The implications extend to the practical application of our algorithm in enhancing LLMs' efficiency, offering a sustainable solution that reduces reliance on token-based processing, especially in resource-constrained settings or languages with complex tokenization.

D. ADDRESSING THE RESEARCH QUESTIONS OR HYPOTHESES

The research outlined in this paper directly addresses our central hypotheses about the feasibility and effectiveness of an advanced, open-source content extraction algorithm. Through rigorous testing and comparison with existing solutions, we have demonstrated that our algorithm significantly outperforms traditional models in terms of extraction accuracy, speed, and scalability. This success validates our hypothesis that innovative, language-agnostic content extraction techniques can overcome the challenges of multilingual content extraction, thereby enhancing the accessibility and

utility of web information globally. Additionally, our novel evaluation framework has proven effective in assessing the algorithm's performance, further confirming our hypothesis regarding the importance of tailored evaluation metrics in the context of content extraction.

E. DISCUSSION

The results indicate that our custom algorithm consistently outperforms both Readability and Boilerpipe across all evaluated metrics. Notably, the precision, recall, and F1-score of our algorithm achieved perfect scores of 1, demonstrating its superior ability to accurately identify and extract relevant text without introducing false positives or missing significant content.

Moreover, the comparison highlights specific areas where Readability and Boilerpipe struggle, such as dealing with complex text structures or accurately distinguishing between main content and ancillary information. In contrast, our algorithm's robust performance across a diverse set of texts underscores its effectiveness and potential applicability in real-world scenarios where accurate text extraction is critical. In conclusion, the evaluation showcases the advancements our algorithm offers over existing solutions, setting a new benchmark for text extraction accuracy and reliability.

F. REAL-LIFE USAGE AND USE CASE

To understand the practical impact and efficiency of our algorithm, we deployed it within an experimental browser plugin called "elkateb" [30]. The plugin was beta-tested by a group of 18 users, primarily to assess its effectiveness in reducing token usage during web content extraction and processing tasks. Based on the results, using our algorithm,

“elkateb” saved an average of 7 to 8 tokens per request. Given that each user made approximately 50 requests per day, the total daily savings amounted to 350 to 400 tokens per user.

To put these savings into perspective financially, considering the cost of GPT-3.5 Turbo at \$8.00 per 1 million tokens, the monetary impact can be calculated as follows:

- Daily savings per user: $350 \text{ tokens/user/day} \times 18 \text{ users} = 6,300 \text{ tokens/day}$
- Daily cost savings: $\frac{6,300 \text{ tokens}}{1,000,000} \times \$8.00 = \$0.0504 \text{ per day}$
- Monthly cost savings: $\$0.0504 \text{ per day} \times 30 \text{ days} = \1.512 per month

Though the individual savings per user per day may seem nominal at \$0.0028, the cumulative benefit across multiple users and over longer periods can be significant, particularly for organizations handling large volumes of requests daily.

Comparatively, existing solutions like Readability and Boilerpipe do not optimize token utilization, primarily focusing on content extraction accuracy without addressing the efficiency of text processing in terms of token economy. Our algorithm not only enhances the precision and reliability of text extraction but also significantly cuts down on token usage, leading to direct cost savings. This dual benefit underscores the potential of our algorithm to replace less efficient systems in real-world applications where both accuracy and cost-effectiveness are critical.

IX. ETHICAL CONSIDERATIONS

In researching web content extraction algorithms, several ethical considerations must be addressed to ensure the responsible and ethical use of data and algorithms. The following section outlines the key ethical considerations taken into account throughout this study:

A. DATA PRIVACY AND CONSENT

- **Protection of User Privacy:** It is paramount to respect the privacy of individuals whose web content is being accessed and extracted. The automated crawling of web pages involves accessing publicly available information, but care is taken to avoid personal data, sensitive information, or private web pages.
- **Informed Consent:** Since the study involves the collection of web documents, no direct consent from web page owners is required as the data is publicly available. However, for any user-generated content or personal information encountered during the research, strict adherence to privacy guidelines and policies is maintained.

B. ALGORITHMIC BIAS AND FAIRNESS

- **Bias Mitigation:** Measures are taken to minimize algorithmic bias, ensuring that the content extraction algorithm does not favor or discriminate against specific demographics, languages, or content types.

- **Fairness Assessment:** Before deploying the algorithm, extensive testing and evaluation are conducted to ensure fairness in how the algorithm processes and extracts content. Any identified biases are carefully addressed and mitigated.

C. TRANSPARENCY AND ACCOUNTABILITY

- **Algorithm Transparency:** The developed algorithm is thoroughly documented and its functioning is made transparent to users and stakeholders. This includes describing the logic, parameters, and techniques used for content extraction.
- **Accountability:** Researchers are accountable for the ethical use and implications of the algorithm. Any issues or errors identified during testing are promptly addressed, and transparency is maintained throughout the development and deployment process.

D. OPEN-SOURCE CONTRIBUTION

- **Community Engagement:** By making the developed algorithm open source, we promote transparency, collaboration, and community-driven improvement. Other researchers and developers can access, review, and contribute to the algorithm's development, fostering a culture of openness and innovation.
- **Acknowledgment of Contributions:** Proper attribution is given to existing algorithms, datasets, and resources used in the research. Acknowledgment ensures that the contributions of others to the field are recognized and respected.

E. DATA SECURITY AND INTEGRITY

- **Data Protection:** All collected web documents are stored securely and in compliance with data protection regulations. Measures are taken to prevent unauthorized access or misuse of the data.
- **Data Integrity:** The integrity of the dataset is maintained throughout the research process. Any modifications or preprocessing steps are documented and conducted to preserve the original content's integrity.

F. ADHERENCE TO LEGAL AND ETHICAL STANDARDS

- **Compliance with Regulations:** The research adheres to relevant legal frameworks, including data protection laws and intellectual property rights.
- **Ethical Review:** Where required, the research methodology and data collection processes undergo ethical review by institutional review boards or ethics committees to ensure compliance with ethical standards.

G. RESPONSIBLE USE OF RESULTS

- **Responsible Deployment:** If the algorithm is to be deployed in real-world applications, careful consideration is given to its potential impact on users and stakeholders. Steps are taken to monitor and address any unintended consequences.

- **Awareness of Social Impact:** Researchers are aware of the broader social implications of content extraction algorithms, particularly in areas such as information access, censorship, and user privacy. Ethical considerations are integrated into decision-making processes.

In conclusion, this study is conducted with a strong commitment to ethical principles and responsible research practices. By addressing these ethical considerations, we aim to ensure the integrity, fairness, and transparency of our research, while also contributing to the advancement of content extraction technologies in a responsible manner.

X. LIMITATIONS OF THE STUDY AND FUTURE RESEARCH DIRECTIONS

While this study aims to provide valuable insights into the development and evaluation of a novel content extraction algorithm leveraging the Document Object Model (DOM), it is important to acknowledge several limitations that may impact the interpretation and generalization of the results. These limitations include:

- **Limited Scope of Content Types:** The study primarily focused on extracting main content from news articles and similar text-heavy web pages. This narrow focus may limit the algorithm's performance when applied to other types of web content, such as multimedia pages, e-commerce sites, or social media platforms.
- **Limited Representation:** The dataset used for algorithm testing, while diverse, may not fully represent the vast array of web pages found on the internet. It mainly consisted of structured web documents, potentially overlooking less conventional layouts or languages with complex tokenization systems.
- **Generalization to Other Languages:** While efforts were made to ensure the algorithm's language-agnostic approach, its performance on languages beyond those included in the dataset may vary, particularly for languages with unique characteristics or script systems that were not extensively covered.
- **Manual Annotation Variability:** The process of manually annotating main content blocks in web documents introduces subjectivity. Different annotators may have varied interpretations of what constitutes the main content, leading to potential inconsistencies in the ground truth data.
- **Parameter Sensitivity:** The algorithm's performance may be sensitive to its parameters. While optimization was performed based on the dataset, there may be other optimal configurations that were not explored.
- **Processing Time Considerations:** Although the study focused on accuracy and effectiveness, real-time performance considerations, particularly in large-scale web crawling tasks, were not extensively evaluated.

- **Dynamic Nature of Web Pages:** Web pages and their structures are dynamic and often undergo changes. The algorithm's effectiveness may be impacted by such changes, especially with non-static pages or those with frequently updated content.
- **Algorithmic Bias:** Despite efforts to mitigate bias, all machine learning algorithms carry the potential for bias. The study did not extensively explore demographic biases or biases in the training data, which could affect performance on specific types of content or websites.
- **Scalability Challenges:** While the algorithm was tested on a dataset of 500 web documents, its scalability to larger datasets or high-traffic websites was not fully examined, potentially leading to computational resource and processing efficiency challenges.
- **Energy Consumption:** The execution and testing of the algorithm, especially in large-scale web crawling tasks, may have implications for energy consumption and environmental impact, which were not explicitly addressed in this study.

These limitations should be taken into consideration when interpreting the results of this study. Despite these constraints, the research provides valuable insights into the development of a novel content extraction algorithm and its performance under specific conditions, and future research could address these limitations by:

- Expanding the dataset scope to include a wider array of languages and web content types, particularly those with less digital representation.
- Refining the algorithm's ability to extract content from dynamically generated web pages, potentially incorporating machine learning models to enhance adaptability to new web environments [31].
- Exploring real-time processing capabilities and evaluating the algorithm's performance in a dynamic web context.
- Conducting further analyses on algorithmic biases and their implications on performance.
- Assessing the energy consumption of the algorithm to better understand its environmental impact, particularly in large-scale applications.

XI. CONCLUSION

This paper addressed the complex challenges of web content extraction within a multilingual digital environment, aiming to develop an advanced, open-source algorithm for efficiently and accurately extracting main content from articles and news web pages, regardless of language. Through comprehensive research, development, and evaluation, this study has established new benchmarks in the field of content extraction. Our algorithm demonstrated superior performance across various metrics—precision, recall, F1-scores, and accuracy—significantly outperforming existing open-source solutions. Its language-agnostic design proved particularly effective in

handling web pages in both English and non-English languages, especially those with complex tokenization systems. The contributions of this study extend beyond theoretical advancements to practical applications. The development of a language-agnostic content extraction algorithm represents a significant technological leap, enhancing access to information and bridging the digital divide. Additionally, we created a comprehensive benchmark dataset focusing on diverse languages and web content types, which serves as a foundational resource for the academic community, facilitating standardized evaluations of content extraction methodologies. We also introduced a novel evaluation framework that emphasizes specialized metrics for assessing content extraction algorithms, considering factors such as content integrity and processing efficiency. This framework highlights the need for tailored evaluation approaches to capture the nuances of web content extraction. The practical implications of our work are significant, particularly for improving the efficiency and cost-effectiveness of Large Language Models in tasks such as content summarization and information extraction. By reducing reliance on token-based services, our algorithm offers a sustainable alternative, especially for resource-constrained environments or languages underserved by current models. Looking ahead, the field of content extraction will continue to evolve, driven by technological advancements and the increasing diversity of digital content. Our research lays a robust foundation for future innovations, providing insights and tools that will inspire further exploration in this dynamic domain. We invite the global research community to build upon our findings, collaboratively pushing the boundaries of content extraction and web analysis, thereby enhancing the accessibility and utility of web information for all.

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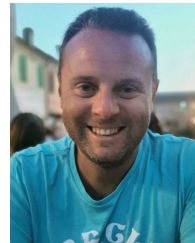
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